PHY202 Practice Exam on chapter(s): 20, 21

1. $I = 786.\,\mathrm{A}\ \because I = \frac{\Delta Q}{\Delta t}$
2. $P = 6.25\,\mathrm{W}\ \because$ Kirchhoff's rules and $P = I^2 R$
3. $P = 7.68\,\mathrm{W}\ \because R_s = R_1 + R_2 + R_3, V = I R, P = I V$
4. $R_{eq} = 0.890\,\Omega\ \because R_s = R_1 + R_2 + R_3, \frac{1}{R_p} = \frac{1}{R_1} + \ldots + \frac{1}{R_n}$
5. $D = 4.88 \times 10^{-5}\,\mathrm{m}\ \because R = \frac{\rho L}{A}$
6. $R = 4.27\,\Omega\ \because V = I R$
7. $I_{RMS} = 650.\,\mathrm{A}\ \because P_{ave} = I_{RMS} V_{RMS}$
8. $E = 1.50 \times 10^4\,\mathrm{J}\ \because P = I^2 R = \frac{E}{t}$
9. $V_0 = 156.\,\mathrm{V}\ \because V_{RMS} = \frac{V_0}{\sqrt{2}}$
10. $Q = 880.\,\mathrm{C}\ \because I = \frac{\Delta Q}{\Delta t}$
11. $E = 3.30 \times 10^3\,\mathrm{J}\ \because P = I V = \frac{E}{t}$
12. $I = 4.03\,\mathrm{A}\ \because \frac{1}{R_p} = \frac{1}{R_1} + \ldots + \frac{1}{R_n}, V = I R$
13. $P = 87.0\,\mathrm{W}\ \because P = I V$
14. $d = 2.03 \times 10^{-6}\,\mathrm{m}\ \because P = \frac{V^2}{R}, R = \frac{\rho L}{A}$ and $A = \pi r^2$
15. $d_{min} = 3.17 \times 10^{-3}\,\mathrm{m}\ \because R = \frac{\rho L}{A}, P_{int} = I^2 R$
16. $V = 6.52\,\mathrm{V}\ \because$ Kirchhoff's rules and $V = I R$
17. $\%\Delta P = -21.3\%\ \because \rho(T) = \rho_0(1 + \alpha \Delta T), R = \frac{\rho L}{A}, P = \frac{V^2}{R}$